

Indian Institute of Technology Guwahati
Probability Theory and Random Processes (MA225)
Problem Set 09

1. Let (X, Y) be bivariate normal such that $Var(X) = Var(Y)$. Show that the two random variables $X + Y$ and $X - Y$ are independent.
2. Let (X, Y) be bivariate normal with parameters $\mu_x = 0, \sigma_x^2 = 1, \mu_y = -1, \sigma_y^2 = 4$, and $\rho = -1/2$.
 - (a) Find $P(X + Y > 0)$.
 - (b) Find the constant a for which $aX + Y$ and $X + 2Y$ are independent.
 - (c) Find $P(X + Y > 0 | 2X - Y = 0)$.
3. Let (X, Y) be bivariate normal with parameters $\mu_x = 0, \sigma_x^2 = 1, \mu_y = 0, \sigma_y^2 = 1$ and correlation coefficient ρ . Using conditional expectation, find $E(X^2 Y^2)$.
4. Let (X, Y) be bivariate normal with parameters $\mu_x = 5, \sigma_x^2 = 1, \mu_y = 10, \sigma_y^2 = 25$ and correlation coefficient ρ , where $\rho > 0$. If it is known that the conditional probability of $Y \in (4, 16)$ given $X = 5$ is 0.954, determine the value of ρ . (Ans: 0.8)
5. Let (X, Y) be bivariate normal with parameters $\mu_x = 0, \sigma_x^2 = 1, \mu_y = 0, \sigma_y^2 = 1, \rho = 0$. Find the real constant c such that

$$P(-c < X < c, -c < Y < c) = 0.95.$$

You can use that $\Phi(2.24) = 0.987$.

6. Assume that the velocity components V_x, V_y, V_z of any molecule of a gas are mutually independent random variables, each being $N(0, \frac{kT}{m})$ where k is Boltzmann's constant, T is the absolute temperature of the gas and m the mass of a molecule. Find the PDF of the velocity $V = \sqrt{V_x^2 + V_y^2 + V_z^2}$.
7. Suppose that the heights of married couples can be explained by a bivariate normal distribution. If the wives have a mean height of 66.8 inches and a standard deviation of 2 inches while the heights of the husbands have a mean of 70 inches and a standard deviation of 2 inches. The correlation between the heights is 0.68. What is the probability that for a randomly selected couple the wife is taller than her husband? Use the fact that $\Phi(2) = 0.977$. (Ans: 0.023)
8. Let X and Y have the bivariate normal distribution. The following facts are known: $\mu_x = 1, \sigma_x = 2$ and the best estimate of Y based on X is given by $3X + 7$. The minimum mean square error is 28. Find μ_y, σ_y and the correlation coefficient ρ between X and Y . (Ans: $\mu_y = 10, \sigma_y = 8, \rho = 3/4$).